

Improving generalized models of forest structure in complex forest types using area- and voxel-based approaches from lidar

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ABSTRACT

Modeling forest attributes using lidar data has been a useful tool for forest management but the need to correlate lidar to ground-based measurements creates challenges to modeling in diverse forest landscapes. Many lidar models have been based on metrics derived from summarizations of individual lidar returns over sample plot areas, but more recently, metrics based on summarization by volumetric pixel (voxel) have shown promise to better characterize forest structure and distinguish between diverse forest types. Voxel-based metrics may improve characterization of leaf area distribution and horizontal forest structure, which could help create general models of forest attributes applicable in complex landscapes composed of many distinct forest types. We modeled wood volume in longleaf pine woodlands and associated forests to compare how area- and voxel- based lidar metrics predicted wood volume in forest type specific and general predictive models. We created four area-based and six voxel-based metrics to fit models of wood volume using a multiplicative power function. We selected models and compared metric importance using AIC and evaluated model performance using cross-validated mean prediction error. We found that one area-based metric and four voxel-based metrics consistently improved model predictions. We suggest that area-based metrics alone may have limitations for characterizing complex forest structure. Area-based summarizations of lidar returns are more heavily influenced by upper canopy returns because lidar returns attenuate below the canopy. By contrast, summarizing lidar returns into a single value per voxel prior to summarization over plots homogenizes point density, giving added weight to sub-canopy returns. Thus voxel-based metrics may be more sensitive to structural variation that may not be adequately captured by area-based metrics alone. This study highlights the potential of voxel-based metrics for characterizing complex forest structure and model generalization capable of accurate forest attribute prediction across diverse forest types.

1. Introduction

Light detection and ranging (lidar) is increasingly applied to improve forest management at various scales. Lidar has been applied to conduct forest inventory (Reutebuch et al., 2005), document stand development (Falkowski et al., 2009), obtain individual tree attributes (Silva et al., 2016), predict wildlife habitat quality (Smart et al., 2012), characterize fuel structure (Loudermilk et al., 2012), and map disturbance and recovery (Hu et al., 2019; Weishampel et al., 2007). The potential to create accurate estimates of tree volume is especially valued for assessment of

forest carbon storage (Houghton et al., 2009). Lidar can provide fine-scale (e.g. < 10 cm) structural information across great extents (e.g. 100 to 1000 km²), providing spatially exhaustive information useful for quantifying landscape-scale processes such as disturbance and succession. In unprocessed form, lidar constitutes points defined in horizontal and vertical space; predictive models are required to transform these point clouds into information on stand structure. Although a variety of lidar models have been used to assess forest structure in many forest types, the value of lidar in diverse natural areas ultimately rests on rigorous examination of model formulations for component ecosystems

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using ground-based measurements.

Traditional lidar models commonly relate area-based summaries of key metrics (e.g., mean, variance, or skewness of lidar return heights) to plot-based measurements of forest structure (Chen et al., 2007; Næsset, 2002; Smart et al., 2012; Wulder et al., 2012). Forest structure at larger scales is then predicted based on these relationships. However, relationships between lidar-derived metrics and forest structure may vary among forest types (Chen, 2013). For example, the relationship between lidar-derived canopy height and carbon (biomass) varies considerably where tree size distributions vary across the landscape (Spriggs et al., 2019), as would be expected when dealing with hardwood- and conifer-dominated forests, or with open- and closed-canopy forest types (Chen et al., 2012; Næsset and Gobakken, 2008). While the development of more generalized models to characterize forest structure across many forest types has shown promise (e.g. Fekety et al., 2018; Tompalski et al., 2019), more work is needed for these models to be transferable to other systems. Additionally, several lidar-derived metrics are consistently informative to forest vegetation models across a wide range of forest types. Height metrics have long been shown to be informative to models in a wide range of forest types. Principal among height metrics is mean canopy height, which has been found to be informative in generalized models of biomass in temperate, boreal, and tropical forests (Asner et al., 2012; Bouvier et al., 2015; Lefsky et al., 2002). Secondly, variance in canopy height has been useful in generalized models of forest attributes (Magnussen et al., 2012). Some generic models with only one or two height-based metrics have yielded accurate predictions, but mean canopy height and height variance are sensitive to differences in lidar instrumentation like the power and divergence of the laser beam (Asner and Mascaro, 2014) and may be less desirable for a generalized model of forest attributes.

Voxel-based models are a newer extension of lidar which may offer advantages over area-based models (Pearse et al., 2019). Voxelization, or the partitioning of point cloud space into a 3-dimensional grid of smaller volumes (voxels), may better account for structural heterogeneity. To predict forest attributes in a wider range of forest types, metrics that describe aspects of vegetation structure other than height are likely to be important. For example, horizontal variability may be more important in low density forests (Hall et al., 2005; Pearse et al., 2019), and metrics that describe vertical canopy leaf area distribution are informative in boreal, temperate, and tropical forests (Knapp et al., 2019; Næsset and Gobakken, 2008). Other lidar metrics can be calculated within volumetric pixels (or “voxels”) which may better capture aspects of horizontal variability and show promise for increasing the accuracy of biomass and volume estimates (Blackburn et al., 2021; Pearse et al., 2019). Recently, Blackburn et al. (2021) demonstrated that voxel-based metrics, when combined with area- and plot-based metrics and machine learning may better characterize forest structure, but it is unclear whether this finding extends to heterogeneous and species diverse forests of the eastern United States. Because they better capture aspects of vertical and horizontal complexity, voxel-based metrics may provide a means to develop a general model for predicting forest structure that is applicable across structurally and compositionally diverse landscapes.

Longleaf pine forests and woodlands of the southeastern U.S. are an ecosystem with high conservation value whose complexity may create challenges for estimating forest structure across large extents. Longleaf pine forests and woodlands support high levels of alpha diversity and endemism (Edwards et al., 2013), but approximately 98% of longleaf pine forests have been lost due to changes in land use (Noss et al., 1996). Though typically dominated by longleaf pine woodlands at the landscape scale, these diverse ecosystems can be composed of many forest types within small areas (e.g., 1000–10,000 ha), owing to topographic and edaphic variability which contribute to high beta diversity (Kirkman et al., 2001, 2004). Longleaf pine-associated communities can include cypress and tupelo swamps, mixed forests dominated by mesic or xeric hardwoods, hardwood hammocks, and stands dominated by

pine species mixtures, among others (Fig. 1; Kirkman et al., 2004). Generating extensive and spatially complete maps of forest structure across multiple forest types is important for understanding ecological processes that operate over large extents (e.g., habitat connectivity and disturbance processes).

In this study, we address two challenges of creating a general model of wood volume in structurally and compositionally complex longleaf pine woodlands and associated forests. We (1) compare the accuracy of forest wood volume estimates generated using area-based approaches (e.g., Bouvier et al., 2015) to those generated from voxel-based approaches (e.g., Blackburn et al., 2021) across multiple longleaf pine-associated forest types; and (2) compare the contributions of area- and voxel-based metrics across both forest type specific and general models of wood volume. Because voxel-based metrics can better account for structural complexity, we hypothesize that voxel approaches offer an improvement over area approaches to estimation of forest structure across forest types. We also hypothesize that voxel-based metrics may be particularly important in generic models of forest structure applicable to multiple forest types.

2. Materials and methods

2.1. Study site

This study was conducted at The Jones Center at Ichauway, an 11,700 ha ecological research facility located on the Southeastern Coastal Plain in Baker County Georgia, USA (31.233° N, 84.467° W). Climate at the site is classified as humid subtropical, with mean annual precipitation of 1360 mm. Summer temperature ranges from 20 to 33 °C, and winter temperature ranges from 3 to 20 °C (NCEI, 2010). Elevation ranges from 21 to 68 m. The majority of landcover in the study area is characterized as longleaf pine forest but differences in hydrology, landscape position, and land use history affect forest species mixtures (Goebel et al., 2001; Kirkman et al., 2004). Riparian and wetland forests can be found along streams which bisect and border the study area (Fig. 2). The karst topography typical of the Southeastern Coastal Plain often includes depressions dominated by mesic oak species and limesink wetlands dominated by pond cypress (*Taxodium ascendens* Brongn.) and blackgum (*Nyssa sylvatica* Marsh.). On drier sandhills and ridges, xeric oaks and other hardwoods account for a larger portion of the species mixture (Goebel et al., 2001; Holland et al., 2019). Pine species other than longleaf pine (*Pinus palustris* Mill.) occur naturally, but commercial forestry activities have resulted in occasional stands dominated by loblolly pine (*Pinus taeda* L.) and slash pine (*Pinus elliottii* Engelm.).

An extraordinarily dense network of inventory plots stratified among plant communities at the site provided forest inventory measurements for our study (Holland et al., 2019; Table 1, Fig. 2). The approach of the study was to generate estimates of wood volume from aerial lidar scans using models that combined area- and/or voxel-based lidar metrics, then assess the accuracy of these estimates by comparing them to forest inventory measurements (Holland et al., 2019). As part of long-term monitoring of forest inventory and biodiversity, in 2002, forests within the study area were classified into nine forest types by photo-interpretation of 30 cm resolution 4-band aerial imagery and screen digitization. Polygons generated in the GIS were field validated and refined. Landcover classification is updated regularly to reflect changes due to growth, management, and disturbances (e.g. planting, harvest, wind damage). Landcover classifications were most recently updated in May 2018 (J. Brock, personal communication; Fig. 2).

2.2. Sample plots

Within each of the nine forest types, we summarized existing forest inventory data from 17.84 m radius circular sampling plots (0.1 ha) established in 2001 as part of a long-term forest monitoring study. These plots were described in detail by Holland et al. (2019). Briefly, a total of



Fig. 1. Nine longleaf pine associated forest types common near Newton, Georgia, USA. Forest types are as follows: A = cypress and tupelo, B = mixed hardwood and pine, C = longleaf pine, D = mixed longleaf pine and hardwood, E = mixed evergreen and deciduous hardwood, F = mixed pine, G = mixed pine and hardwood, H = planted pine, I = upland hardwood.

864 sample plots were distributed across the study area using a random tessellation stratified design divided into four panels, resulting in all plots being re-measured over a four-year interval (Holland et al., 2019). All plots were located using GPS and marked with a permanent monument. For each plot, we obtained data on all trees ≥ 10 cm DBH (diameter at breast height or 1.37 m above the ground), including species, DBH, and height. We calculated wood volume following Holland et al. (2019) using allometric equations and species-specific coefficients (Sheffield and Schweitzer, 2002; Woodall et al., 2010). Sample plots were classified into nine forest types (Tables 1 and 2). Sample plots in non-forested landcover classes were omitted from modeling and analyses. Omission of an additional 16 sample plots resulted from limited lidar coverage in 2017 (Fig. 2). A total of 653 sample plots were included in modeling and analyses.

Forest structure in the nine forest types spanned a large range of stand density and species composition, and tree form (Table 2). Pine-dominated forest types in the study area can be classified as savannas or woodlands (Scholes and Archer, 1997) and typically had lower basal area with smaller crowns and higher gap fractions than hardwood dominated forest types. Structure in cypress dominated wetland forest was unique with high density and crown areas and gap fraction intermediate between pine-dominated and hardwood-dominated forest types.

2.3. Lidar data collection and processing

Lidar data are available in the study area through the National Ecological Observatory Network's (NEON) Airborne Observation Platform (Kampe et al., 2010). Discrete return lidar data were collected using a Riegl LMS-Q780 lidar scanner flown approximately 1000 m above ground level. Data meet USGS quality level 2 standards with an average point density of $6.9 \text{ points m}^{-2}$ (Heidemann, 2012). Data were collected near maximum greenness conditions in September of the years 2014, 2016, 2017, 2018, and 2019 (Fig. 2). Raw point data was processed by NEON (e.g. calibration, orthorectification, classification) following published protocols and standards (NEON, 2022).

To create both area-based and voxel-based lidar metrics, we used the R packages *lidR* (Roussel et al., 2020) and *sf* (Pebesma, 2018). We normalized lidar point clouds to height above ground, clipped them to sample plot areas, and removed ground returns. We used methodology from Bouvier et al. (2015) to summarize area-based metrics within each plot boundary. To calculate mean canopy height (μ_{ch}) and canopy height variance (σ_{ch}), we created canopy height models for each plot using first returns and summarized them by mean and variance, respectively at a resolution of 1 m^2 . To describe horizontal forest structure, we calculated gap fraction (P_g) by taking the ratio of the number of lidar returns below 2 m in height to the total number of returns. To describe vertical arrangement of vegetation within a plot, we calculated the coefficient of

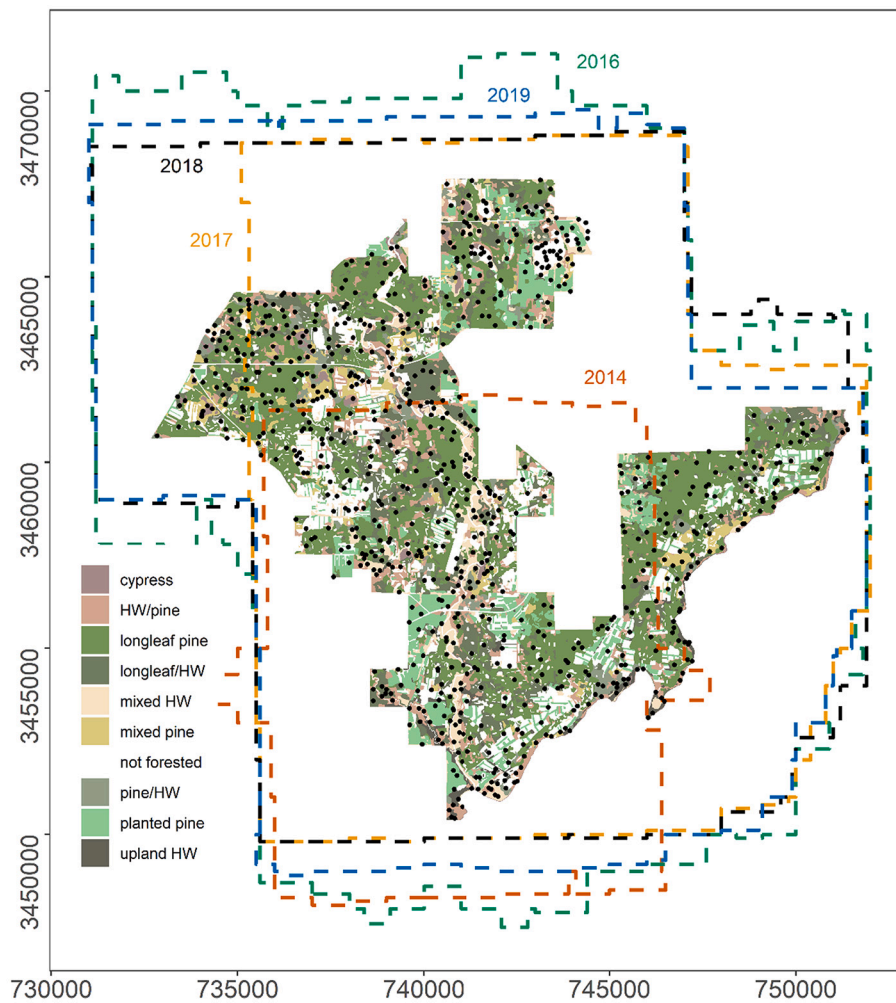


Fig. 2. Plot location, lidar coverage, and forest cover types used for wood volume models in longleaf pine associated forests near Newton, Georgia, USA. Circular, black symbols show sample plot locations, and dashed lines indicate extent of lidar coverage each year by colour as follow: orange = 2014, green = 2016, yellow = 2017, black = 2018, and blue = 2019. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Number of sample plots in each forest type (n) and percentage of stems by functional group in each forest type for sample plots in longleaf pine associated forests near Newton, Georgia, USA.

Forest Type	No. Of plots	Deciduous	Evergreen oak	Longleaf pine	Other pine	Wetland	Other
Cypress	21	9.1	6.8	0.2	4.9	66.8	12.1
HW/pine	80	23.3	36.4	12.2	7.1	9.0	12.0
Longleaf pine	230	5.8	6.4	79.8	4.4	1.1	2.6
Longleaf/HW	85	21.7	19.0	53.8	2.1	1.2	2.3
Mixed HW	69	26.8	35.6	4.5	1.6	9.9	21.6
Mixed pine	41	6.4	8.9	23.8	54.0	3.6	3.3
Pine/HW	45	13.3	25.9	16.6	25.2	12.0	7.0
Planted pine	60	1.6	3.2	51.6	40.5	0.0	3.3
Upland HW	22	39.2	20.5	5.2	2.8	10.2	22.0

variation in leaf area density (cv_{lad}). We estimated leaf area density in 1 m height intervals from 2 m to the maximum return height using the ratio of the number of returns that reached the interval height layer to the number of returns below that layer, then took the coefficient of variation of leaf area densities (Bouvier et al., 2015).

In addition, we calculated five voxel-based lidar metrics identified by Blackburn et al. (2021) as the most informative to models that describe aspects of southwestern forest structure. A detailed description of the calculation of these metrics can be found in Blackburn et al. (2021). For all voxel-based metrics, we first subset point clouds into 1 m³ voxels, counted the number of returns within each voxel, and calculated the

mean and median of the height of returns within each voxel. These voxel values were then summarized over plot volumes. To describe vertical forest structure, we calculated the mean and median of voxel height means by plot ($\mu\mu_{Ht}$ and $\eta\mu_{Ht}$, respectively). To describe 3-dimensional heterogeneity in the distribution of vegetation in a plot, we calculated the coefficient of variation and mean of the ratio of returns above each voxel to the total number of returns in the plot (CV_{fr} and μ_{fr}). Finally, as a descriptor of both vertical and horizontal variability, we calculated the coefficient of variation of the total number of returns below each voxel (CV_p).

Table 2

Structural attributes in nine forest types common in longleaf pine associated forests near Newton, Georgia, USA. Values are means with standard deviation (in parentheses) across field plots by forest type.

Forest Type	Density (trees ha ⁻¹)	Height (m)	BA (m ² ha ⁻¹)	Crown Area (m ²)	Gap Fraction (m ² m ⁻²)
Cypress	719.8 (363.9)	16.7 (4.6)	31 (16)	61.1 (22.4)	0.2 (0.1)
HW/pine	340.7 (174.4)	20.5 (4.9)	22.4 (11)	95.1 (42.8)	0.1 (0.1)
Longleaf pine	207.7 (116.5)	19.3 (5.1)	13.4 (6.8)	54.8 (25.2)	0.2 (0.1)
Longleaf/ HW	209.1 (121.8)	16.9 (4.8)	12.5 (6.3)	59.1 (27.4)	0.2 (0.1)
Mixed HW	389.6 (140.4)	20 (4.1)	25.4 (10)	88.7 (29.2)	0.1 (0.1)
Mixed pine	195.2 (92)	19.3 (5.6)	13.9 (6.4)	63.3 (32.7)	0.2 (0.1)
Pine/HW	264.5 (149.2)	19.5 (5.3)	17.4 (9)	65.3 (26.2)	0.2 (0.1)
Planted pine	697.3 (532.9)	13.9 (4.2)	12.3 (7.5)	34.2 (10.2)	0.2 (0.1)
Upland HW	331.9 (108.1)	20.8 (4.5)	21.3 (7.6)	86.3 (29.5)	0.1 (0.1)

2.4. Statistical analysis

We fit models using maximum likelihood estimation that included all combinations of the ten candidate lidar metrics mentioned above. We used simulated annealing as a global optimization algorithm to find model parameter values that best agreed with observations, and modeled error with a normal probability density function (Bolker, 2008). We fit 1023 models for each of the nine forest types and for all forest types combined for a total of 10,230 unique models. We used the multiplicative power model form to predict wood volume because it has been widely used to predict forest attributes from remotely sensed data (Knapp et al., 2019; Lim and Treitz, 2004; Næsset and Gobakken, 2008):

$$Volume = \beta_0 \cdot P_1^{\beta_1} \cdot P_2^{\beta_2} \cdot \dots \cdot P_n^{\beta_n}$$

Where P_i represents a single predictor variable i (i.e. lidar-derived metric) and β_i represents an estimated model coefficient for P_i . Due to non-linearity in response variables, we log-transformed lidar metric predictors and observed wood volumes prior to modeling.

We calculated coefficient of determination (R^2) and Akaike's information criterion corrected for small sample size (AICc) for each model (Burnham and Anderson, 2002). We chose to use AICc to compare model performance because it is a measure of how well a model will predict new data whereas R^2 is a measure of how well the model fits data being modeled. We identified the lowest AICc-score models that included only area-based metrics and the lowest AICc-score model that included voxel-based metrics by forest type and compared them to each other and to null models (models that only included an intercept) using Δ AICc. We ranked the models by forest type from low to high AICc, then grouped the model with the lowest AICc score and all models within 2 AICc units of that model together by forest type (top-models). Models within 2 AICc units of the best model can be regarded as having substantial support as the most likely model (Burnham and Anderson, 2002). To assess how informative metrics were to models of wood volume, we tallied metric inclusion in top-models by forest type and calculated the mean Δ AICc that resulted when a metric was removed from each top-model in which it was included.

To assess prediction accuracy, we calculated symmetric median absolute percentage error (SMAPE) for predictions made using leave one out cross validation (LOOCV). At each sample plot, i , SMAPE is defined as

$$SMAPE = \frac{1}{N} \sum_{i=1}^N \text{median} \left(\frac{200|\hat{y}_i - y_i|}{|\hat{y}_i + y_i|} \right)$$

where N is the number of sample plots, and $\hat{y}_i$ and y_i are the predicted and observed wood volumes at the sample plot. Measures of error based on the mean of a series of model predictions, such as we made using LOOCV, can be biased toward models that underpredict (Hyndman and Koehler, 2006; Tofallis, 2015). Therefore, we chose SMAPE, which gives a robust error measure suitable for comparison of competing models (Armstrong and Collopy, 1992; Blackburn et al., 2021). Finally, to assess bias in our three modeling approaches (forest type specific, area-based general, and area-and-voxel-based general), we fit linear regression models of predicted versus observed wood volume by forest type and used one-sample t -tests to determine if regression slopes significantly differed from 1.

3. Results

We found that models based on area-based metrics consistently predicted volume better than null models (intercept only models). The best (lowest AICc) models that included only area-based metrics had significantly more support than null models with Δ AICc ≥ 2 in all forest types (Table 3, Burnham and Anderson, 2002). The variation in wood volume explained by the best area-based models ranged from 33% in the mixed hardwood forest type to 80% in cypress dominated forests. Only models for mixed-hardwood and planted pine forests explained <50% of the variation (Table 3).

We found that inclusion of voxel-based metrics alongside area-based metrics further improved prediction in many forest types even after being penalized for including additional parameters (Table 3). The top generic models, which were trained on data from all forest types with a wider range of structural variability, each included four metrics. Eight of the nine top forest type specific models, which were trained on data from a single forest type, included fewer metrics than top generic models and had smaller sample sizes. Voxel-based metrics were included in the top generic model and in seven of the nine forest type-specific models (Table 3). Of those, the generic, HW/pine, longleaf pine, and planted pine models had significantly more support than models including only area-based metrics. The best models in mixed hardwood and upland hardwood forest types did not include any voxel-based metrics, and except for those two forest types, the variation in wood volume explained by models that included voxel-based metrics was always greater than in models that included only area-based metrics.

We identified one area-based metric and four voxel-based metrics that were consistently important for predicting wood volume in many forest types (Table 4). Removal of μ_{CH} , μ_{FR} , CV_p , μ_{HT} , or $\eta\mu_{HT}$ from top-models resulted in models with significantly less support (≥ 2 AICc higher on average) than the models from which they were removed. The height metric, μ_{CH} , was the most consistently important. It was included in 59% of top-models overall and its removal resulted in a mean increase of 20 AICc units. The voxel-based metric, μ_{HT} , was the next most consistently important metric and was included in one or more top-models in seven of the ten forest types and in 38% of top-models overall (Table 4). Its removal from top-models resulted in a mean increase of 24 AICc units. In addition, the mean increase in AICc that resulted from its removal in all forests and longleaf/HW top-models was greater than that of μ_{CH} . The next most included metric, CV_p , is a voxel-based metric that was included in top-models for all types, cypress dominated, mixed hardwood/pine, longleaf pine, and planted pine forests, and in 42% of top-models overall. Its removal from top-models resulted in a mean increase of 7 AICc units. Its inclusion in every top-model in all forests and in 95% of longleaf pine forests, and the relatively large increase in AICc that resulted from its removal from top-models suggests it is an important descriptor of tall open canopies, with minimal midstory like those common in longleaf pine dominated forests (Fig. 2C,F).

Table 3

Comparison of models of wood volume in nine forest types common in longleaf pine associated forests near Newton, Georgia, USA. Parameter estimates are shown for metrics that were included in each model. The best area-based (a) and area-voxel-based (v) models are listed. In forest types where the best model did not include voxel-based metrics, only the area-based model is listed. Bold font and grey shading indicate the lowest AICc model for each forest type. Abbreviations are: HW = hardwood, LL = longleaf pine, Mix. = mixed, P = pine, and Upl. = upland.

Forest type	$\Delta AICc$	R^2	int	μ_{ch}	σ_{ch}	P_f	CV_{lad}	CV_{FR}	μ_{FR}	CV_P	$\mu\mu_{Ht}$	$\eta\mu_{Ht}$
All (a)	20.64	0.60		1.63	-0.22	0.13	-0.39					
All (v)	0.00	0.64			-0.25		-0.20			-0.76	1.95	
Cypress (a)	0.18	0.80	0.34	2.19								
Cypress (v)	0.00	0.81								-1.30	1.78	
HW/pine (a)	2.39	0.54	0.62	1.41								
HW/pine (v)	0.00	0.60		1.48						-0.77		
Longleaf (a)	9.27	0.69	0.40	2.02		0.28						
Longleaf (v)	0.00	0.76		1.84		0.17	-0.30			-0.80		
LL/HW (a)	1.16	0.62	0.78	1.24								
LL/HW (v)	0.00	0.64						-0.35			1.37	
Mix. HW (a)	0.00	0.33		1.68			-0.69					
Mix. pine (a)	0.00	0.54		1.05								
Pine/HW (a)	0.69	0.55		0.99								
Pine/HW (v)	0.00	0.62		2.25						-1.11		
Planted P. (a)	3.39	0.42		0.98								
Planted P. (v)	0.00	0.55								-1.08	1.67	
Upl. HW (a)	1.29	0.53		0.92								
Upl. HW (v)	0.00	0.64										0.72

int = intercept, μ_{ch} = mean canopy height, σ_{ch} = variance in canopy height, P_f = gap fraction, CV_{lad} = coefficient of variation of leaf area density, CV_{FR} = coefficient of variation of the ratio of returns above a voxel to the total number of returns, μ_{FR} = mean of the ratio of returns above a voxel to the total number of returns, CV_P = coefficient of variation of the number of returns below a voxel, $\mu\mu_{Ht}$ = mean of the mean height of returns within a voxel, $\eta\mu_{Ht}$ = median of the mean height of returns within a voxel.

We found that the generic models (both area- and voxel-based) generally had lower error relative to forest type-specific models (Fig. 3). In every forest type, the generic area-based model (G) and generic-area-voxel-based models (GV) predicted wood volume with

equal or significantly less ($p \leq 0.05$) error than forest type specific models (S; Fig. 3). Prediction error in generic models was significantly lower than in forest type specific models in longleaf pine, longleaf/HW, mixed HW, mixed pine, and pine/HW forest types. In longleaf/HW

Table 4

Metric importance in wood volume models in nine forest types common in longleaf pine associated forests near Newton, Georgia, USA. We calculated metric importance by forest type using the model with the lowest AICc along with any models that were within two AICc units of that model. Values are mean $\Delta AICc$ when the metric was removed from each model it was included in, and the proportion of models that each metric appeared in (in parentheses). Bold font indicates $\Delta AICc > 2$ units. Abbreviation: HW = hardwood.

Forest type	n	int	μ_{ch}	σ_{ch}	P _f	CV _{lad}	CV _{FR}	μ_{FR}	CV _P	$\mu\mu_{Ht}$	$\eta\mu_{Ht}$
all	14	-1.4 (0.07)	38.6 (0.57)	3.1 (0.71)	2.2 (0.43)	2.9 (1.00)	-1.6 (0.07)	-2.0 (0.07)	25.6 (1.00)	93.4 (0.86)	-1.1 (0.29)
cypress	25	1.9 (0.16)	6.6 (0.32)	0.8 (0.12)	NA (0.00)	1.7 (0.16)	2.3 (0.16)	5.8 (0.24)	2.3 (0.16)	5.8 (0.28)	6.2 (0.32)
HW/pine	6	NA (0.00)	15.6 (1.00)	NA (0.00)	-1.8 (0.17)	NA (0.00)	4.0 (0.17)	5.1 (0.33)	5.2 (0.50)	NA (0.00)	-1.0 (0.33)
longleaf pine	22	1.6 (0.18)	44.4 (0.82)	NA (0.00)	2.1 (0.45)	1.2 (0.27)	2.8 (0.27)	-0.5 (0.05)	17.5 (0.95)	39.2 (0.50)	-0.0 (0.45)
longleaf/HW	44	1.2 (0.34)	16.2 (0.34)	-1.0 (0.11)	-0.2 (0.30)	-1.5 (0.09)	2.9 (0.57)	20.1 (0.45)	1.5 (0.32)	23.2 (0.34)	-0.5 (0.14)
mixed HW	14	0.7 (0.21)	7.5 (0.86)	-2.0 (0.07)	-1.2 (0.07)	4.9 (0.71)	1.0 (0.14)	1.7 (0.57)	NA (0.00)	3.9 (0.36)	-0.3 (0.07)
mixed pine	14	0.1 (0.07)	15.5 (0.36)	-0.6 (0.07)	-1.9 (0.07)	-0.8 (0.21)	0.3 (0.07)	NA (0.00)	-0.6 (0.21)	9.7 (0.57)	2.8 (0.21)
pine/HW	10	-0.5 (0.10)	23.4 (0.90)	-0.1 (0.10)	-0.7 (0.10)	-0.3 (0.10)	-0.4 (0.10)	0.7 (0.10)	0.1 (0.10)	-0.1 (0.10)	20.9 (0.10)
planted pine	20	0.7 (0.35)	27.5 (0.20)	-1.5 (0.10)	-0.5 (0.30)	-0.3 (0.25)	1.1 (0.45)	-1.0 (0.25)	4.7 (1.00)	18.8 (0.80)	NA (0.00)
upland HW	2	NA (0.00)	5.8 (0.50)	NA (0.00)	NA (0.00)	NA (0.00)	NA (0.00)	NA (0.00)	NA (0.00)	NA (0.00)	7.1 (0.50)
overall	171	0.5 (0.15)	20.1 (0.59)	-0.2 (0.13)	-0.2 (0.19)	1.0 (0.28)	1.4 (0.20)	3.7 (0.21)	7.0 (0.42)	24.2 (0.38)	3.8 (0.24)

n = number of models within 2 AICc of the best model, int = intercept, μ_{ch} = mean canopy height, σ_{ch} = variance in canopy height, P_f = gap fraction, CV_{lad} = coefficient of variation of leaf area density, CV_{FR} = coefficient of variation of the ratio of returns above a voxel to the total number of returns, μ_{FR} = mean of the ratio of returns above a voxel to the total number of returns, CV_P = coefficient of variation of the number of returns below a voxel, $\mu\mu_{Ht}$ = mean of the mean height of returns within a voxel, $\eta\mu_{Ht}$ = median of the mean height of returns within a voxel.

forests, prediction error from area-voxel-based models was significantly lower than from area-based models in longleaf/HW forests. There were no other statistically significant differences between the area-voxel-based models and area-based models, but prediction error from area-voxel-based models was lower than from area-based models in every forest type except upland HW where the opposite was true.

Plots of predicted wood volume compared to observed values

indicate that slopes and intercepts varied among models by forest type, but there were a few notable patterns of prediction bias in our estimates of wood volume (Table 5, Fig. 4). Area-based- and area-voxel-based-general models had slopes <1 in hardwood dominated forests and slopes >1 in longleaf pine and cypress dominated forests. The number of forest types with slope values for observed versus predicted wood volume that were statistically different from 1 was higher in general models

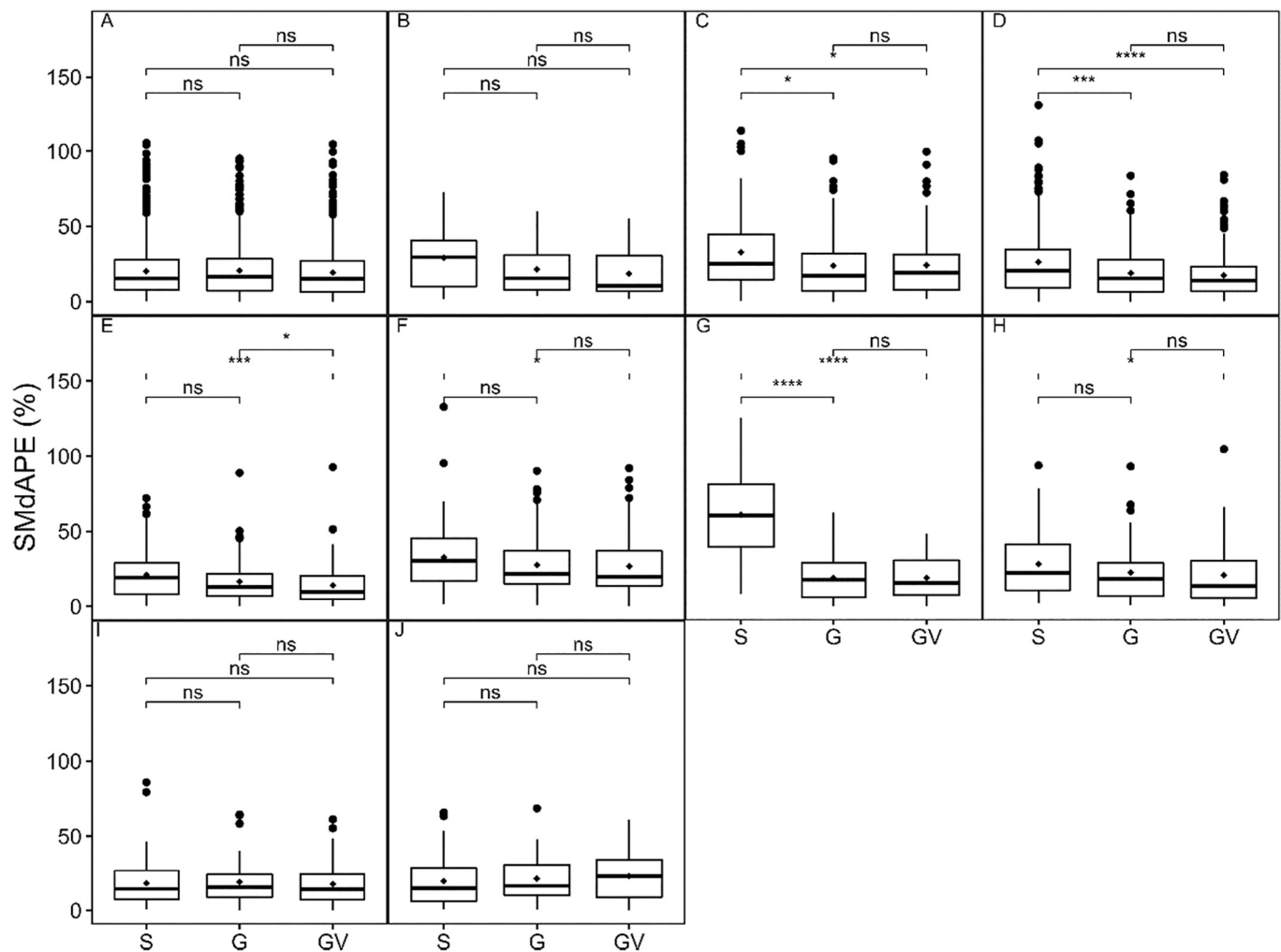


Fig. 3. Boxplots of symmetric median absolute percentage error of observed vs. predicted wood volume in nine common longleaf pine associated forest types near Newton, Georgia, USA. Diamonds indicate mean values. Brackets indicate differences between pairs of modeling approaches (Wilcoxon test). Stars or text above brackets indicate p -values: ns = not significant, * = 0.05, ** = 0.01, *** = 0.001, **** < 0.0001. Abbreviations for modeling approaches are: S = forest type specific using area-based and voxel metrics; G = generic using area-based metrics; GV = generic using area-based and voxel-based metrics. Panels are: A = all, B = cypress, C = HW/pine, D = longleaf pine, E = longleaf/HW, F = mixed HW, G = mixed pine, H = pine/HW, I = planted pine, and J = upland HW.

than in forest type specific models ($S = 2$, $G = 3$, and $GV = 4$; Table 5). Forest type specific and both general models underpredicted at high wood volume in longleaf pine forests and overpredicted at high wood volume in planted pine stands. General models overpredicted at low wood volume in hardwood dominated forests (HW/pine, mixed HW, upland HW). Modeled wood volumes were consistently the most biased in planted pine forests (slopes = 0.49, 0.45, 0.47 for forest type-specific, area-based-general, and area-voxel-based-general models, respectively), and overpredicted low wood volumes and underpredicted high wood volumes (Table 5).

4. Discussion

Overall, we found that the inclusion of voxel-based metrics can improve predictions of wood volume and may be useful for reducing prediction error in open canopy forest types. Interestingly only one area-based metric related to canopy height (μ_{CH}) consistently improved predictions of wood volume. On the other hand, four voxel-based metrics (μ_{FR} , CV_p , $\mu\mu_{HT}$, and $\eta\mu_{HT}$) consistently improved predictions. Though models that included voxel-based metrics seldom produced significantly lower prediction errors, many voxel-based metrics were included in high proportions of top-models and their removal often had stronger effects on $\Delta AICc$ values than the removal of area-based metrics. Given the high

structural variability among forest types included in this study (Table 2, Fig. 2), this may indicate that voxel-based metrics are more sensitive to both horizontal and vertical variation in forest structure, attributes that may not be adequately captured by area-based metrics alone. Voxel-based metrics may hold particular promise for predicting forest attributes in forest ecosystems with high structural and compositional variability.

Our results support the hypothesis that voxel-based metrics are more sensitive for capturing variation in horizontal forest structure and leaf arrangement compared to area-based metrics (Pearse et al., 2019), and they further suggest that voxel-based metrics may also be more sensitive to complex variation in vertical forest structure. Though mean canopy height was important in top-models for every forest type, it was often absent from the lowest $AICc$ models. The fact that several voxel-based metrics, including CV_p and $\mu\mu_{HT}$, and to a lesser extent, μ_{FR} , CV_{FR} , and $\eta\mu_{HT}$, were included in high proportions of top-models and generally reduced $AICc$ indicates that they may be quantifying aspects of structural variability not captured by area-based metrics such as vertical openness beneath the canopy, strata complexity, and/or height differentiation within dense forest types.

Longleaf pine woodlands are typified by high crowns and nearly empty midstories (Fig. 2C, D, F, G). Midstories in mixed and longleaf pine associated hardwood-dominated forest types are often occupied by

Table 5

Fit statistics for linear regressions of observed versus predicted wood volumes for forest type-specific (S), area-based-general (G), and area-voxel-based-general (GV) models of wood volume in nine forest types common in longleaf pine associated forests near Newton, Georgia, USA. Bold font and grey background indicate slopes that are significantly different from 1 ($p = 0.05$).

Forest Type	S			G			GV		
	Int	slope	R ²	Int	slope	R ²	Int	slope	R ²
cypress	8.98	1.02	0.71	8.54	1.25	0.68	-14.70	1.32	0.76
HW/pine	13.87	1.01	0.39	56.36	0.92	0.34	54.06	0.93	0.34
longleaf pine	-4.36	1.07	0.69	4.10	1.06	0.58	-9.09	1.15	0.64
longleaf/HW	-16.41	1.20	0.58	21.12	0.90	0.56	2.73	1.03	0.61
mixed HW	100.98	0.65	0.21	73.91	0.86	0.26	97.76	0.69	0.20
mixed pine	-3.25	1.16	0.51	20.37	0.73	0.50	-9.20	0.87	0.57
pine/HW	63.20	0.72	0.33	66.45	0.67	0.31	70.47	0.66	0.31
planted pine	56.74	0.49	0.31	62.73	0.45	0.28	58.10	0.47	0.33
upland HW	-41.11	1.32	0.38	40.37	0.79	0.37	38.53	0.78	0.29

dense midstory trees, and trees in suppressed canopy positions that may not be adequately characterized by metrics such as mean canopy height which are heavily influenced by returns in the upper canopy (Fig. 2E, I). Voxel-based metrics may better capture the complex and variable arrangement of leaf area in diverse forests by summarizing lidar returns by voxel prior to voxel summarization into forest metrics, effectively increasing the weight given to midstory returns in calculations. For instance, $\mu\mu_{Ht}$ and $\eta\mu_{Ht}$ calculate the mean height of returns within each voxel before summarizing the mean and median heights, which reduces the number of returns that occupy a voxel to a single value, in effect homogenizing point density and giving equal weight to all occupied voxels. This emphasizes vertical and horizontal forest structure throughout the point cloud, especially for ALS where lidar returns attenuate below the canopy. Similarly, the voxel-based measurements of CV_{fr} , μ_{FR} , and CV_p may provide better measures of vertical and horizontal forest structural variability than area-based metrics. These metrics summarize lidar returns in the voxels above or below a voxel, respectively, and thus provide information on vertical structure and light environment. The vertical series of voxels that share the same x, y coordinates form a stack of voxels that provide a measure of vertical variability, and because these voxel stacks are in a grid that covers the area of a plot, they provide a measure of horizontal variability as well. Summarization of these vertically-informed voxels over the plot area incorporates horizontal variation and provides a measure of canopy openness. Indeed, the large proportions of top-models that included CV_p and $\mu\mu_{Ht}$ coupled with relatively large $\Delta AICc$ values suggests that these metrics were important in distinguishing a wide range of forest structures (i.e., differences in initial planting density or tree spatial pattern or where size distributions or species composition were more variable) and may be important predictors when developing generalized models.

The trend of lower prediction error in general models compared to forest type-specific models indicates potential benefits of modeling over a wider range of variability but may illustrate the challenges of modeling forest attributes with remotely sensed data. We selected lidar metrics

using a priori knowledge; an approach suggested to alleviate overfitting problems associated with stepwise selection methods (Chen et al., 2007). Even so, we sought to measure metric importance by ranking our models using AICc, and only later comparing the lowest AICc models by prediction error. Ranking models using prediction error generated from LOOCV could help identify metrics important in forest attribute modeling; however, this would require conducting LOOCV on 10,230 models which would be a computationally expensive task. The lack of significant differences in prediction error between area-based and area-voxel-based generic models also suggests that ranking models using LOOCV-generated prediction error is unlikely to yield different results.

Despite the clear importance of voxel metrics (Table 4), prediction errors from general models that did not include voxel metrics were often not significantly different from those that included voxel-based metrics (Fig. 3). This is in contrast with Blackburn et al. (2021) who found that wood volume models for *Pinus ponderosa*-dominated forests in the southwestern United States with voxel-based metrics decreased prediction error by 7% compared to models with only area-based metrics. However, prediction error in our best general model of wood volume with voxel-based metrics was similar to prediction error in their best model (20.5 and 20.8 SMdAPE, respectively). Our model results also are comparable to results from general wood volume modeling in mountainous areas in France (Bouvier et al., 2015; $R^2 = 0.63$, mean prediction error = -22.86%). Pearse et al. (2019) reported higher R^2 values (0.73 to 0.75) in planted *Pinus radiata* stands in southeastern Australia, but our models spanned a greater range of forest types and likely encompassed a greater range of structural variability. This may suggest that forest attribute modeling is likely not solely constrained by the combination of metrics used to model forest attributes. Our forest attribute modeling as well as most previous modeling efforts were based on regression models which are constrained by a chosen mathematical form, in our case, a multiplicative power model (Coops et al., 2021). A few researchers have had success modeling forest attributes using machine learning techniques (Blackburn et al., 2021; Cartus et al., 2012; Luther et al., 2019).

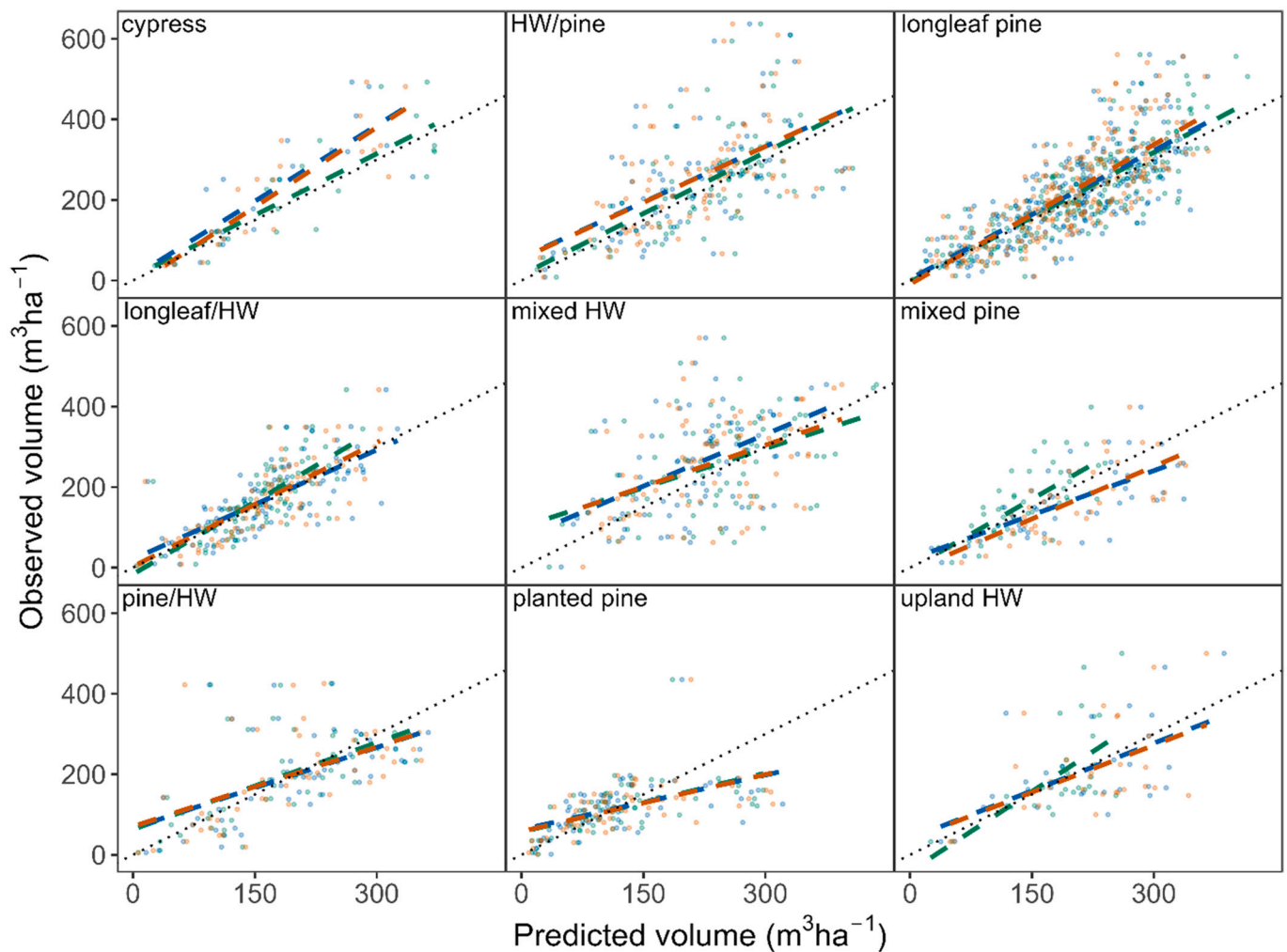


Fig. 4. Observed versus predicted wood volume and linear regressions for forest type specific models (green points and lines), the area-based-general model (blue points and lines), and the area-voxel-based-general model (orange points and lines) in nine forest types on the southeastern coastal plain, USA. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The benefit of using a machine learning approach is that the data and model fit are not constrained by a mathematically rigid form. This could be important for creating a highly generalizable model because the relationships between forest attributes and lidar metrics, upon which regression models rely, may vary substantially by forest type.

Largely building on the previous works of [Bouvier et al. \(2015\)](#) and [Blackburn et al. \(2021\)](#), we set out to explore the importance of area-based and voxel-based prediction attributes for developing general models of forest biomass in structurally and compositionally complex forests. We found that voxel-based models continue to show improvements over area-based models. Our results also show that model development approaches including voxel-based prediction attributes could further improve our ability to predict diverse forest attributes under a wide range of canopy structures and for forest types not encountered (and thus not sampled). While continued exploration into building and validating generalized models of forest structure from airborne lidar is needed, our results suggest that generalized models of forest structure transferred to novel conditions may be able to provide accurate predictions in data-poor situations or under conditions not empirically observed in previous modeling efforts, thus providing great potential to inform and improve management decisions.

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CRediT authorship contribution statement

Andrew W. Whelan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Jeffery B. Cannon:** Conceptualization, Methodology, Investigation, Resources, Writing – original draft, Writing – review & editing. **Seth W. Bigelow:** Conceptualization, Investigation, Resources, Writing – review & editing. **Brandon T. Rutledge:** Validation, Investigation, Data curation, Writing – review & editing. **Andrew J. Sánchez Meador:** Methodology, Investigation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Lidar point cloud data used in this work are freely available from <https://data.neonscience.org>, and field inventory and reference volume data will be made available by the authors upon request.

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